

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi

CS406: Mid-semester Examination
25th January 2023

Max. marks: 20

Time: 1 hr

Q 1. Find the real root of the equation $3x = \cos x + 1$, using Newton-Raphson method.

4

Q 2. Solve the following system of equations using the Gauss-Jordan method

4

$$\begin{aligned}x + y + z &= 1 \\4x + 3y - z &= 6 \\3x + 5y + 3z &= 4\end{aligned}$$

Q 3. Find the smallest positive root of $x^2 - 5x + 1 = 0$, correct to four decimal places

4

Q 4. Test the consistency of the following system of equations

$$\begin{aligned}x + 10y - z &= 3 \\2x + 3y + 20z &= 7 \\9x + 22y + 79z &= 45.\end{aligned}$$

4

Q 5. Using Lagrange interpolation, find the unique polynomial $P(x)$ of degree 2 or less such that $P(1) = 1$, $P(3) = 27$, $P(4) = 64$.

4

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
Mid-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
8th October 2024

Time: 2 hours

Total Marks: 20

Answer all questions.

Q 1. Find the interval in which the positive roots of the equation $x - e^{-x} = 0$ lie. Find the smallest positive roots of the equation using Regula-Falsi method. 5

Q 2. Find the solution of the system of equations

$$4x_1 + x_2 + x_3 = 4$$

$$x_1 + 4x_2 - 2x_3 = 4$$

$$3x_1 + 2x_2 - 4x_3 = 6$$

using the Gauss elimination method. 5

Q 3. Using Lagrange's formula, find the interpolating polynomial $P(x)$ of degree 2 or less that takes the following values. 10

$$x_0 = 1, P(x_0) = 1$$

$$x_1 = 3, P(x_1) = 27$$

$$x_2 = 4, P(x_2) = 64$$

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
Mid-semester Examination- II

CS 406 - Numerical Methods
MCA 1st Semester
27 November 2024

Time: 2 hrs

Max. Marks: 20

Q 1. Use Newton's forward difference formula to find the unique polynomial $P(x)$ such that $P(1) = 1$, $P(2) = 0$, $P(3) = 27$ and $P(4) = 64$ (5)

Q 2. Compute the following integral using Trapezoidal rule for $p = 1$, and number of points = 3 (5)

$$I_p = \int_0^1 \frac{x^p dx}{x^3 + 10}$$

Q 3. Compute approximation to $y(0.6)$ (4 decimals) of the IVP (10)

$$\begin{aligned} y' &= x(x+y) - 2 \\ y(0) &= 2 \end{aligned}$$

Using Euler's method with step size $h = 0.2$

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
End-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
25th September 2025

Time: 2 hours

Total Marks: 20

Answer all questions

Q 1. What is linear independence? Explain how the concept of linear independence is important for solving a system of linear equations. 5

Q 2. Find the inverse of the coefficient matrix of the system of linear equations

$$x + y + z = 1$$

$$4x + 3y - z = 6$$

$$3x + 5y + 3z = 4$$

using the Gauss-Jordan method and then solve the system. 10

Q 3. Given the equation,

$$x^2 + 5 = e^x$$

first, find the interval in which the positive roots of the equation lie and then, using Newton-Raphson method, find the smallest positive roots of the equation. 5

School of Computer and Systems Sciences
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New Delhi

Mid-semester - 2 Examination
Numerical Methods (CS-406)
11th November 2025

Time: 2 hours

Total Marks: 20

Answer all questions.

Q 1. For the given the data below, find the polynomial $P(x)$ such that $P(x_i) = y_i \forall i = 0, 1, 2$ and then, find its first derivative, $P'(x)$ (8)

$$x_0 = 0, y_0 = 0$$

$$x_1 = 1, y_1 = 1$$

$$x_2 = 3, y_2 = 0$$

Q 2. Find the approximate value of

$$I = \int_0^1 f(x) dx$$

where

$$f(x) = \frac{1}{1+x}$$

using the trapezium rule with 4 equal subintervals. (8)

Q 3. What is the role of interpolation in animation. Explain in detail.
(4)